

PATTERN MINING II

SESSION 7
COMP6140001 – DATA MINING

SUBJECT MATTER EXPERT
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LEARNING OBJECTIVE

L02 : to apply various data mining techniques



OUTLINE

- Frequent itemset mining methods: Frequent pattern growth (FP-growth)
- Pattern evaluation Methods

FREQUENT ITEMSET MINING METHODS : FREQUENT PATTERN GROWTH (FP-GROWTH)



FREQUENT ITEMSET MINING METHODS : FREQUENT PATTERN GROWTH (FP-GROWTH)

Mining frequent patterns without candidate generation

- Bottlenecks of the Apriori approach
 - Breadth-first (i.e., level-wise) search
 - Candidate generation and test
 - Often generates a huge number of candidates
- The FPGrowth Approach (J. Han, J. Pei, and Y. Yin, SIGMOD' 00)
 - Depth-first search
 - Avoid explicit candidate generation
- Major philosophy: Grow long patterns from short ones using local frequent items only
 - “abc” is a frequent pattern
 - Get all transactions having “abc”, i.e., project DB on abc: DB|abc
 - “d” is a local frequent item in DB|abc → abcd is a frequent pattern



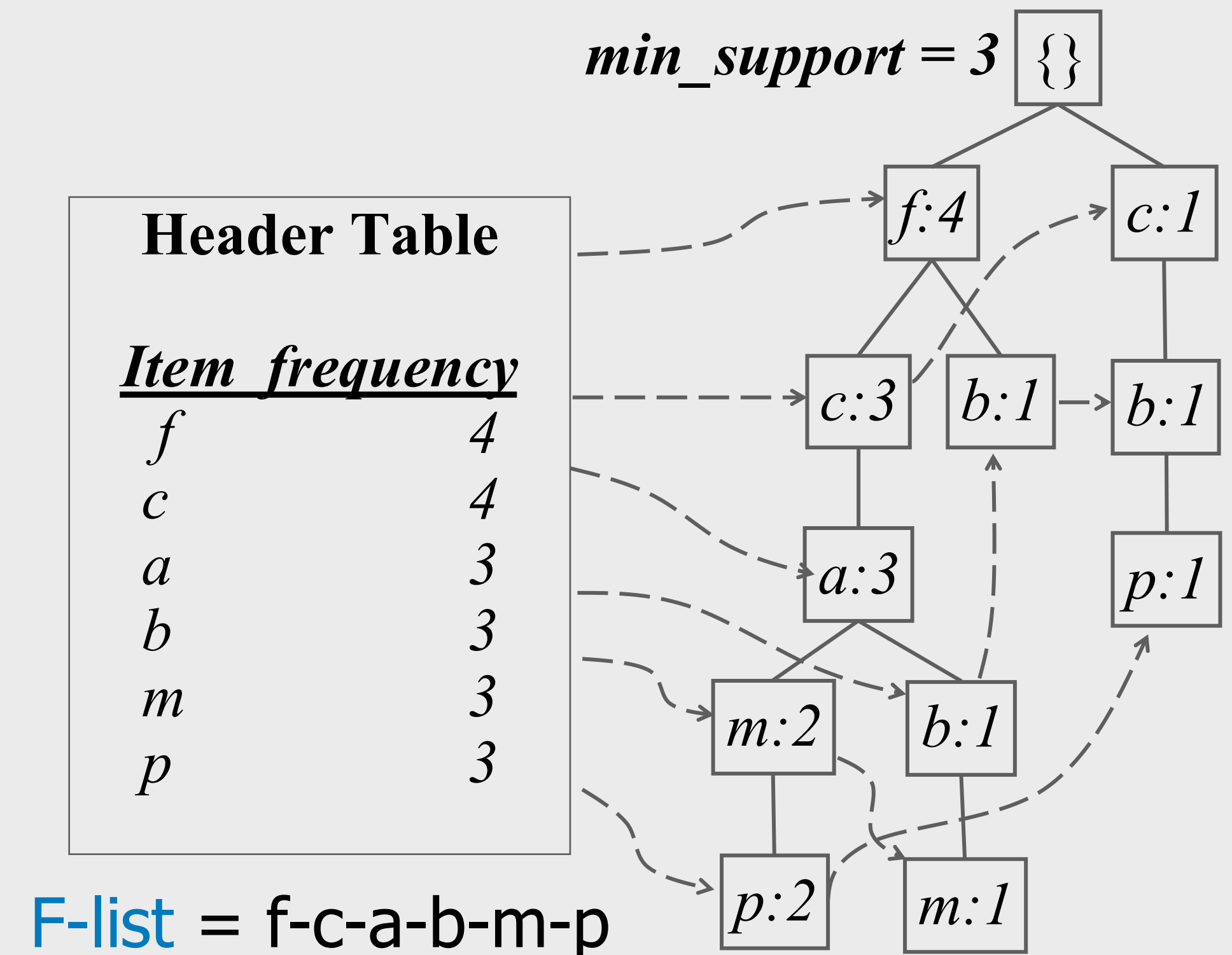
PATTERN-GROWTH APPROACH (2)

Construct FP-tree from a transaction database

1. Scan DB once, find frequent 1-itemset (single item pattern)
2. Sort frequent items in frequency descending order, f-list
3. Scan DB again, construct FP-tree

<i>TID</i>	<i>Items bought</i>	<i>(ordered) frequent items</i>
100	{f, a, c, d, g, i, m, p}	{f, c, a, m, p}
200	{a, b, c, f, l, m, o}	{f, c, a, b, m}
300	{b, f, h, j, o, w}	{f, b}
400	{b, c, k, s, p}	{c, b, p}
500	{a, f, c, e, l, p, m, n}	{f, c, a, m, p}

Header Table	
<i>Item frequency</i>	
<i>f</i>	4
<i>c</i>	4
<i>a</i>	3
<i>b</i>	3
<i>m</i>	3
<i>p</i>	3





PATTERN-GROWTH APPROACH (3)

Partition patterns and databases

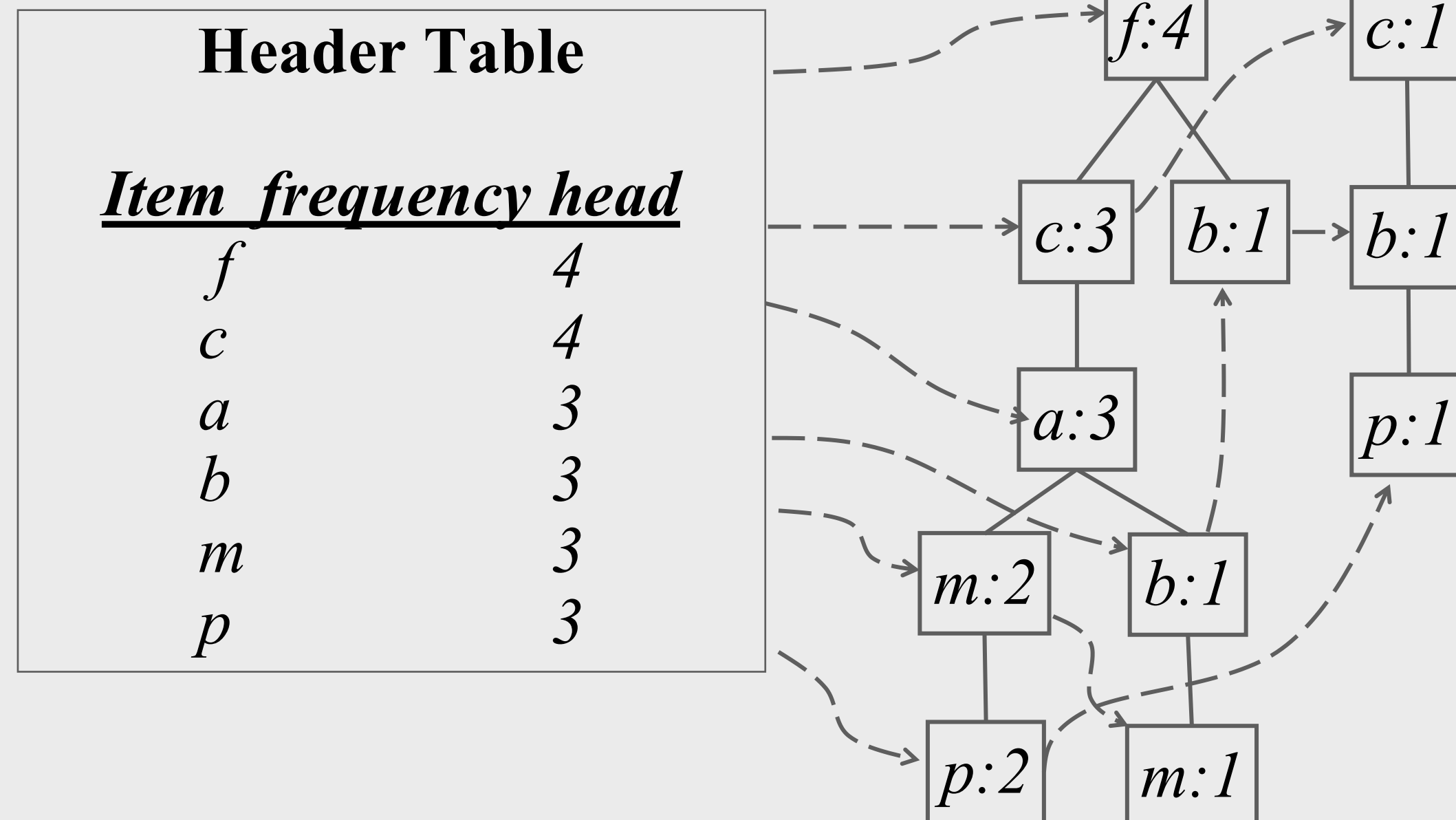
- Frequent patterns can be partitioned into subsets according to f-list
 - F-list = f-c-a-b-m-p
 - Patterns containing p
 - Patterns having m but no p
 - ...
 - Patterns having c but no a nor b, m, p
 - Pattern f
- Completeness and non-redundancy



PATTERN-GROWTH APPROACH (4)

Find patterns having P from P-conditional database

- Starting at the frequent item header table in the FP-tree
- Traverse the FP-tree by following the link of each frequent item p
- Accumulate all of **transformed prefix paths** of item p to form p's conditional pattern base



Conditional pattern bases

item *cond. pattern base*

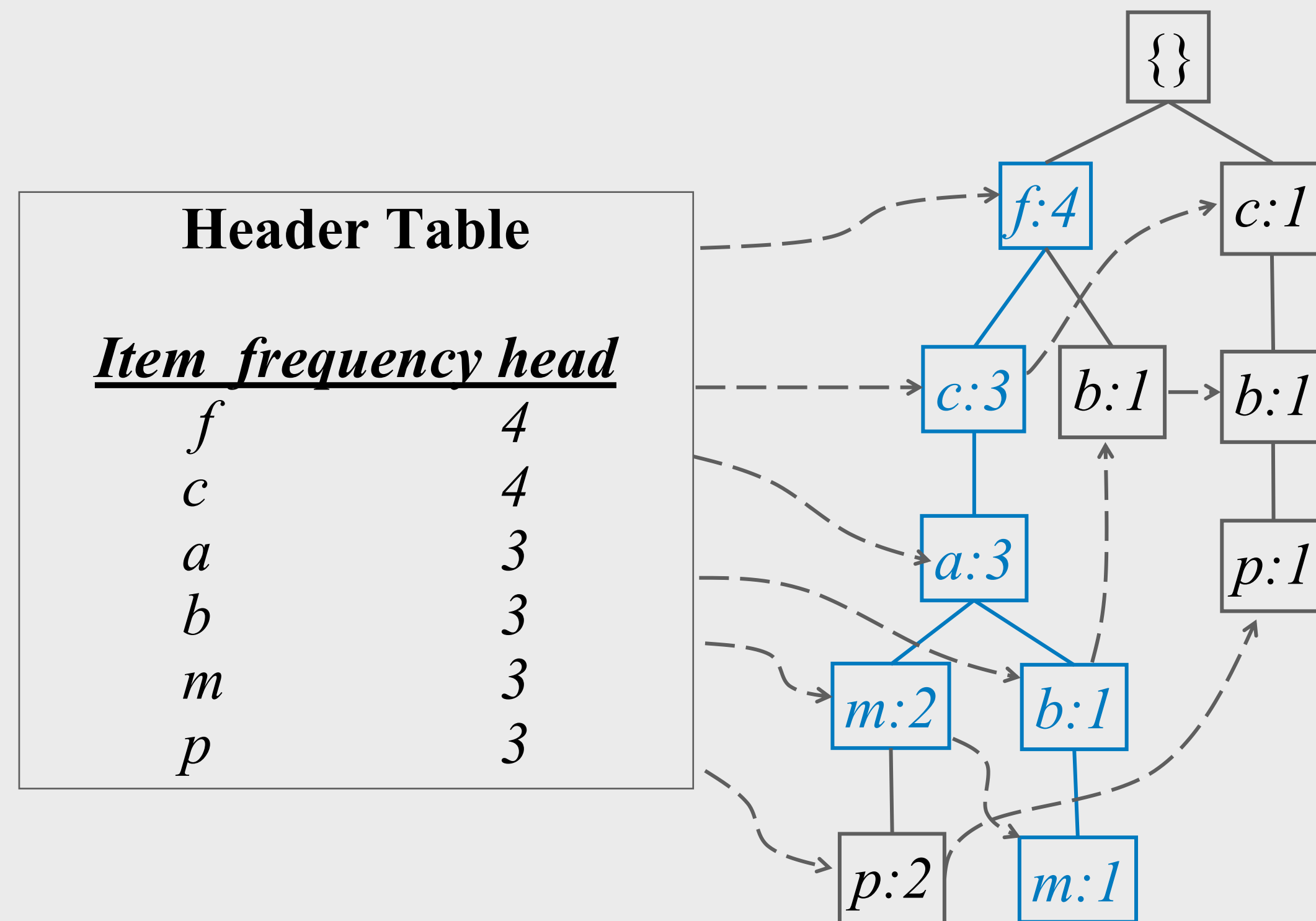
<i>c</i>	<i>f:3</i>
<i>a</i>	<i>fc:3</i>
<i>b</i>	<i>fca:1, f:1, c:1</i>
<i>m</i>	<i>fca:2, fcab:1</i>
<i>p</i>	<i>fcam:2, cb:1</i>



PATTERN-GROWTH APPROACH (5)

From conditional pattern-bases to conditional FP-trees

- For each pattern-base
 - Accumulate the count for each item in the base
 - Construct the FP-tree for the frequent items of the pattern base



m-conditional pattern base:
fca:2, fcab:1

$\{ \}$
|
f:3
|
c:3
|
a:3



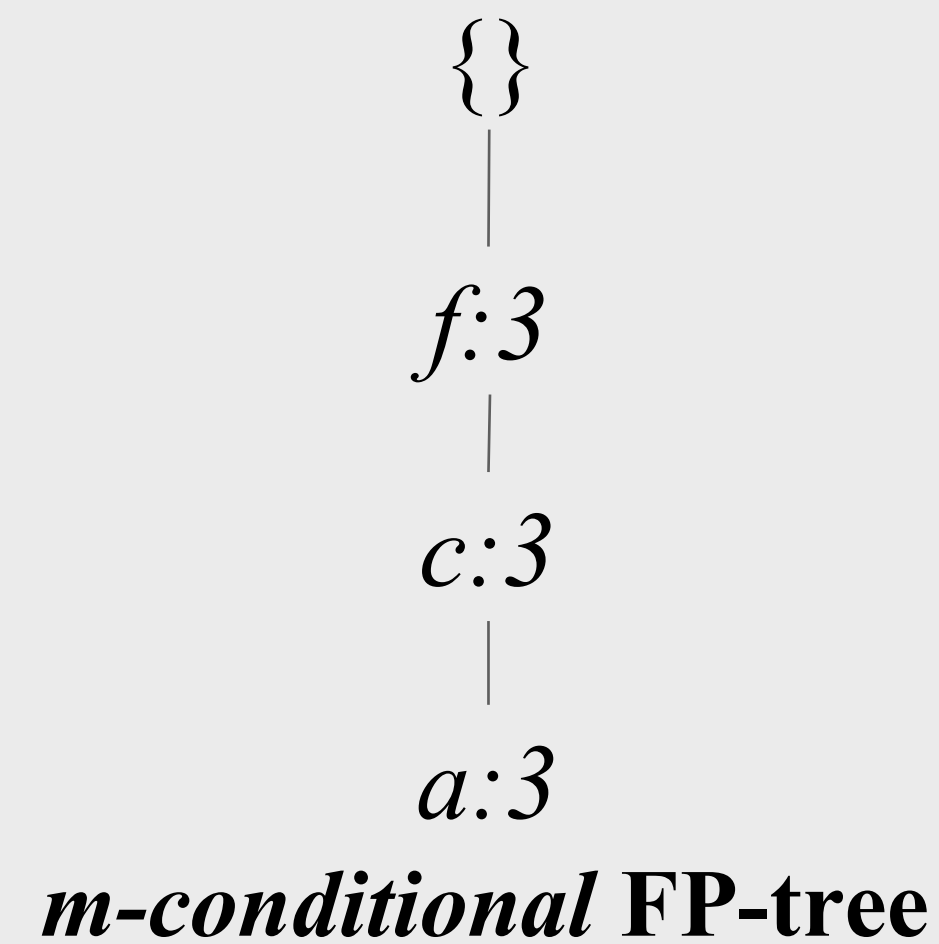
**All frequent
patterns relate to *m***
m,
fm, cm, am,
fcm, fam, cam,
fcam

m-conditional FP-tree



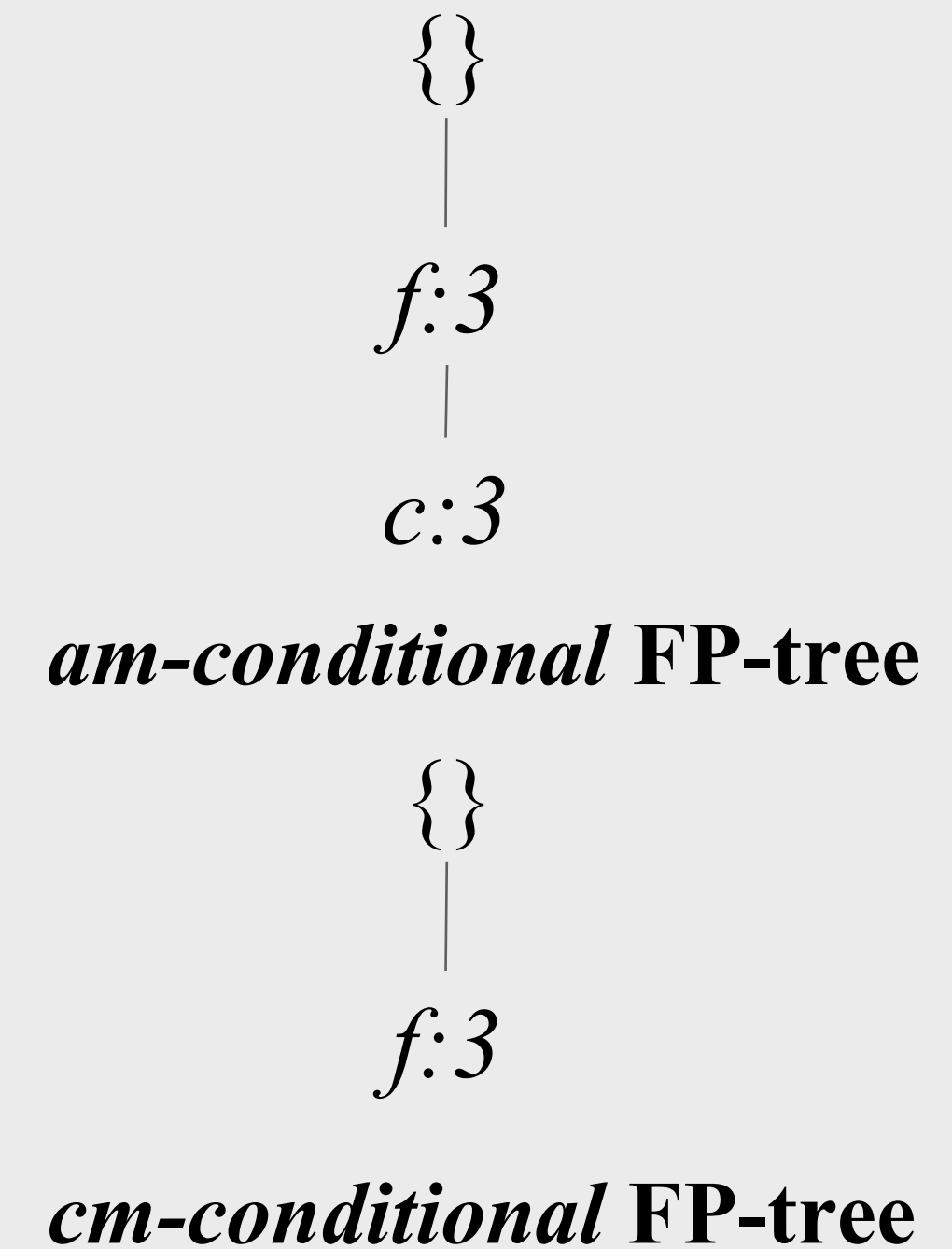
PATTERN-GROWTH APPROACH (6)

Recursion: mining each conditional FP-tree

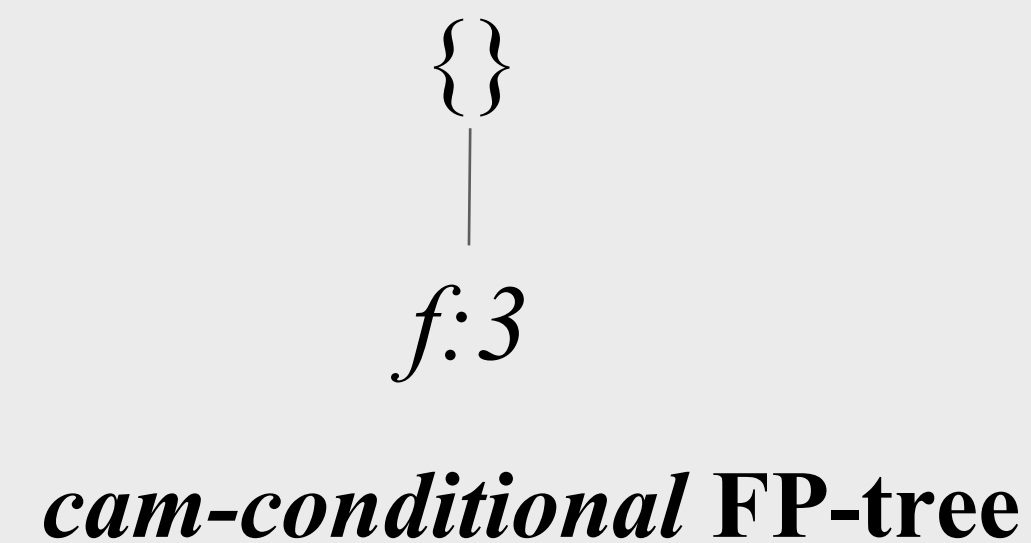


Cond. pattern base of "am": (fc:3)

Cond. pattern base of "cm": (f:3)



Cond. pattern base of "cam": (f:3)

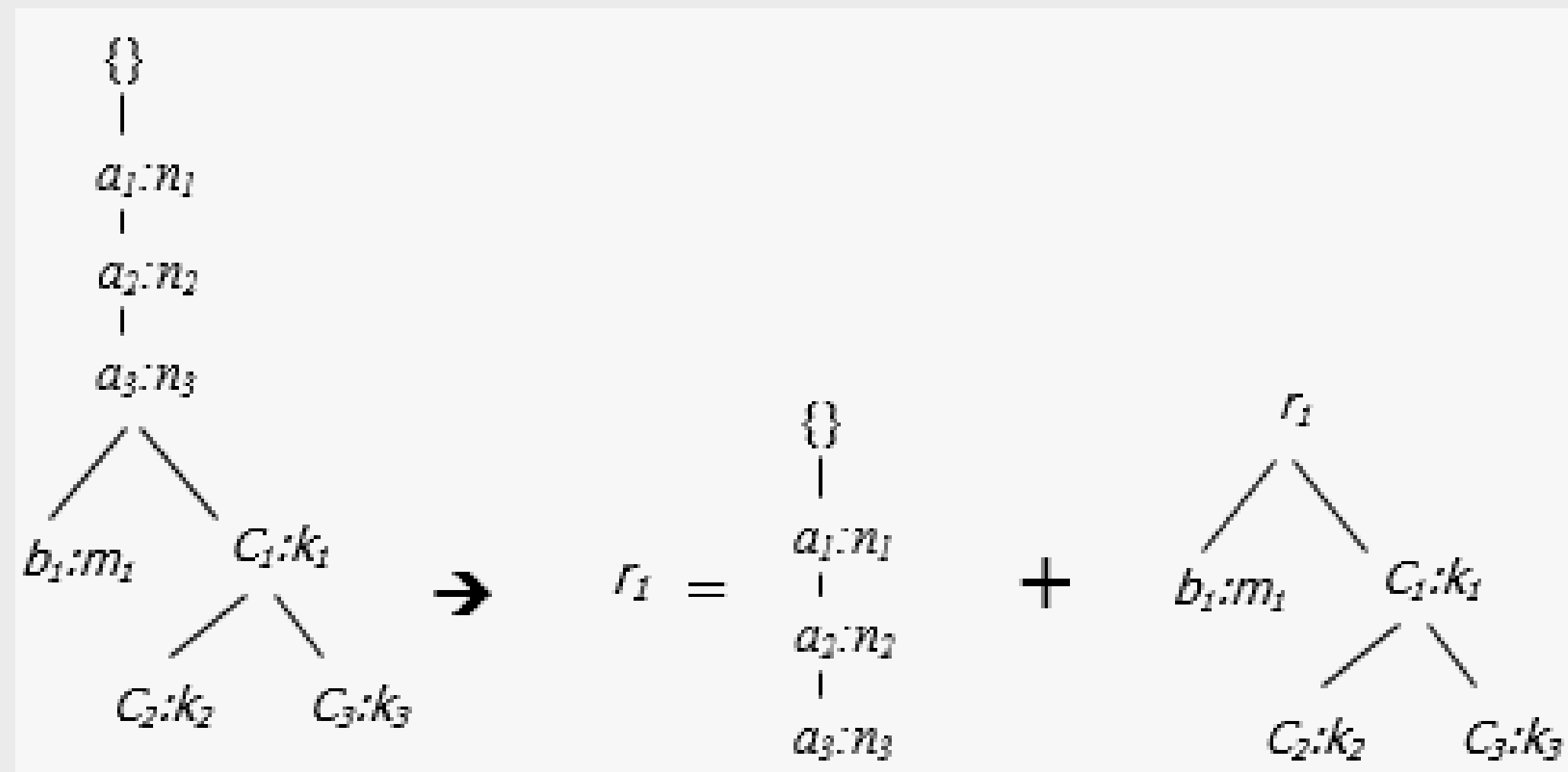




PATTERN-GROWTH APPROACH (7)

A special case: single prefix path in FP-tree

- Suppose a (conditional) FP-tree T has a shared single prefix-path P
- Mining can be decomposed into two parts
 - Reduction of the single prefix path into one node
 - Concatenation of the mining results of the two parts





PATTERN-GROWTH APPROACH (8)

Benefits of the FP-tree structure

- **Completeness**
 - Preserve complete information for frequent pattern mining
 - Never break a long pattern of any transaction
- **Compactness**
 - Reduce irrelevant info—infrequent items are gone
 - Items in frequency descending order: the more frequently occurring, the more likely to be shared
 - Never be larger than the original database (not count node-links and the count field)



PATTERN-GROWTH APPROACH (9)

The frequent pattern growth mining method

- Idea: Frequent pattern growth
 - Recursively grow frequent patterns by pattern and database partition
- Method
 - For each frequent item, construct its conditional pattern-base, and then its conditional FP-tree
 - Repeat the process on each newly created conditional FP-tree
 - Until the resulting FP-tree is empty, or it contains only one path—single path will generate all the combinations of its sub-paths, each of which is a frequent pattern



PATTERN-GROWTH APPROACH (10)

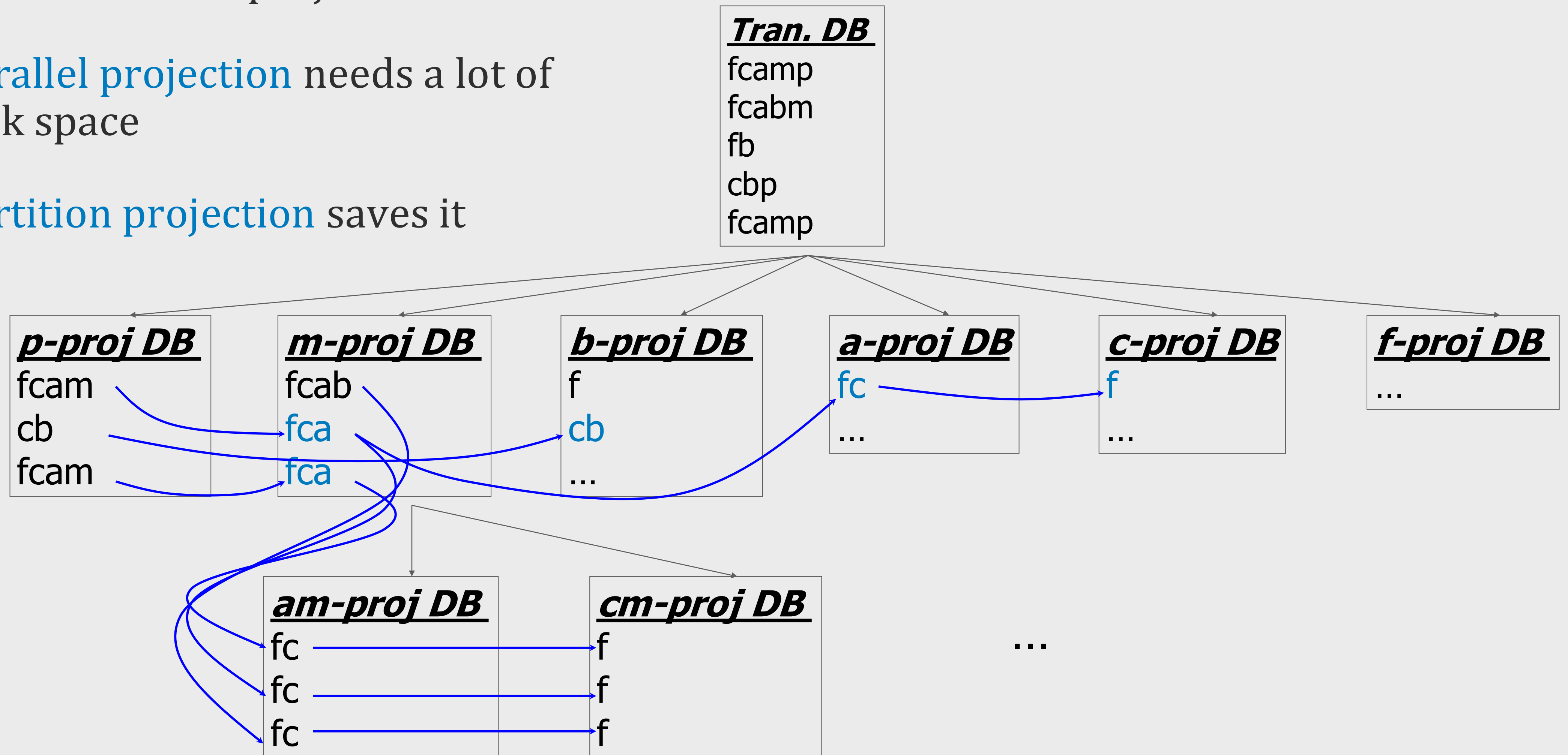
Scaling FP-growth by database projection

- What about if FP-tree cannot fit in memory?
 - DB projection
- First partition a database into a set of projected DBs
- Then construct and mine FP-tree for each projected DB
 - Parallel projection vs. partition projection techniques
 - Parallel projection
 - Project the DB in parallel for each frequent item
 - Parallel projection is space costly
 - All the partitions can be processed in parallel
 - Partition projection
 - Partition the DB based on the ordered frequent items
 - Passing the unprocessed parts to the subsequent partitions



PATTERN-GROWTH APPROACH (11)

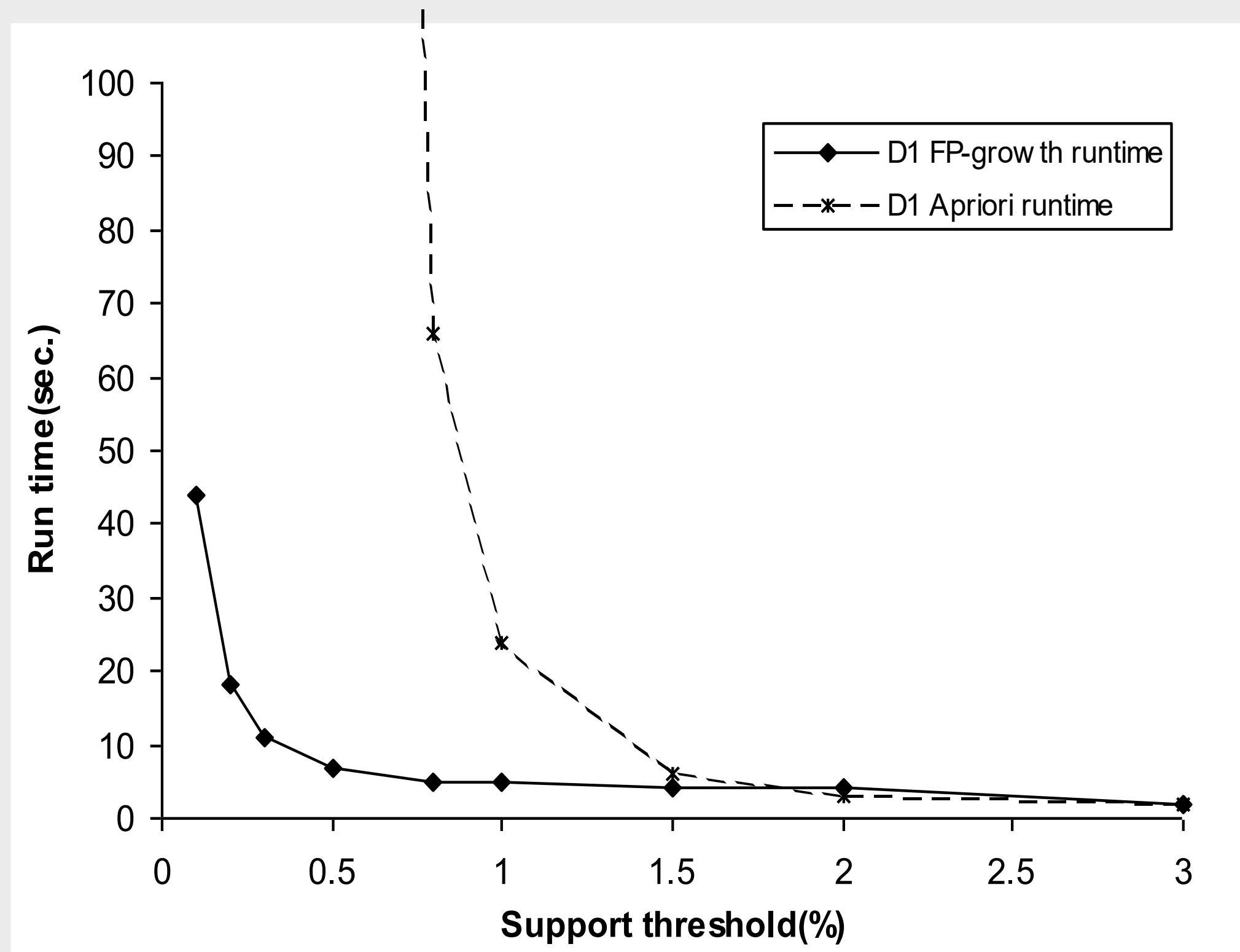
- Partition-based projection
- Parallel projection needs a lot of disk space
- Partition projection saves it



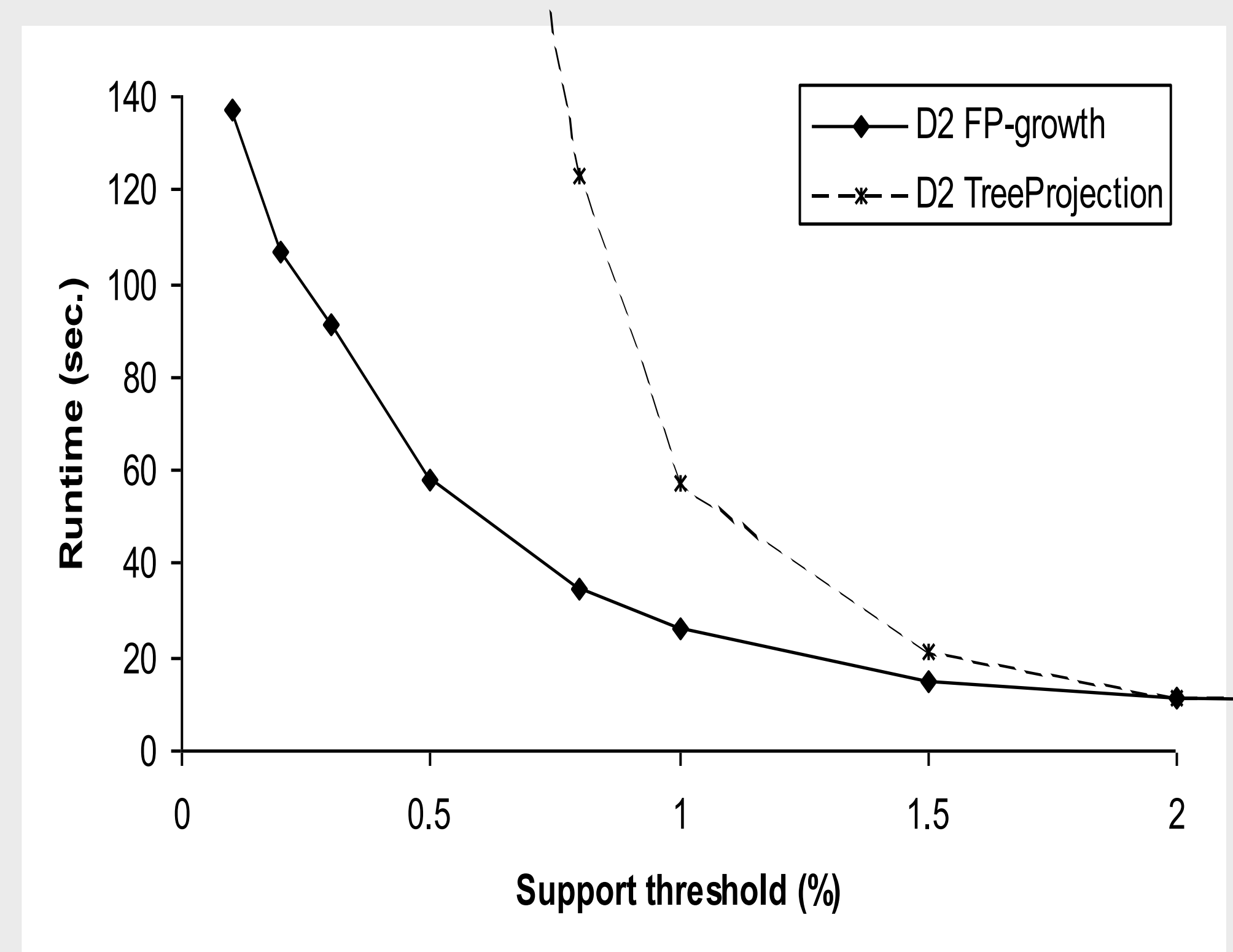


PATTERN-GROWTH APPROACH (12)

Performance of FPGrowth in large datasets



FP-Growth vs. Apriori



FP-Growth vs. Tree-Projection



PATTERN-GROWTH APPROACH (13)

Advantages of the pattern growth approach

- Divide-and-conquer:
 - Decompose both the mining task and DB according to the frequent patterns obtained so far
 - Lead to focused search of smaller databases
- Other factors
 - No candidate generation, no candidate test
 - Compressed database: FP-tree structure
 - No repeated scan of entire database
 - Basic ops: counting local freq items and building sub FP-tree, no pattern search and matching
- A good open-source implementation and refinement of FPGrowth
 - FPGrowth+ (Grahne and J. Zhu, FIMI'03)



MINING BY EXPLORING VERTICAL DATA FORMAT

CHARM

- Vertical format: $t(AB) = \{T_{11}, T_{25}, \dots\}$
 - tid-list: list of trans.-ids containing an itemset
- Deriving frequent patterns based on vertical intersections
 - $t(X) = t(Y)$: X and Y always happen together
 - $t(X) \subset t(Y)$: transaction having X always has Y
- Using **diffset** to accelerate mining
 - Only keep track of differences of tids
 - $t(X) = \{T_1, T_2, T_3\}$, $t(XY) = \{T_1, T_3\}$
 - $\text{Diffset}(XY, X) = \{T_2\}$
- Eclat (Zaki et al. @KDD'97)
- Mining Closed patterns using vertical format: CHARM (Zaki & Hsiao@SDM'02)



MINING FREQUENT CLOSED PATTERNS

CLOSET

- Flist: list of all frequent items in support ascending order **Min_sup=2**
 - Flist: d-a-f-e-c
- Divide search space
 - Patterns having d
 - Patterns having d but no a, etc.
- Find frequent closed pattern recursively
 - Every transaction having d also has cfa → cfad is a frequent closed pattern
- J. Pei, J. Han & R. Mao. "CLOSET: An Efficient Algorithm for Mining Frequent Closed Itemsets", DMKD'00.

TID	Items
10	a, c, d, e, f
20	a, b, e
30	c, e, f
40	a, c, d, f
50	c, e, f



MINING FREQUENT CLOSED PATTERNS (2)

CLOSET+ (by Pattern-Growth)

- Itemset merging: if Y appears in every occurrence of X , then Y is merged with X
- Sub-itemset pruning: if $Y \supset X$, and $\text{sup}(X) = \text{sup}(Y)$, X and all of X 's descendants in the set enumeration tree can be pruned
- Hybrid tree projection
 - Bottom-up physical tree-projection
 - Top-down pseudo tree-projection
- Item skipping: if a local frequent item has the same support in several header tables at different levels, one can prune it from the header table at higher levels
- Efficient subset checking



MINING FREQUENT CLOSED PATTERNS (3)

MaxMiner (mining max-patterns)

- 1st scan: find frequent items

- A, B, C, D, E

- 2nd scan: find support for

- AB, AC, AD, AE, **ABCDE**

- BC, BD, BE, **BCDE**

- CD, CE, **CDE**, DE

Tid	Items
10	A, B, C, D, E
20	B, C, D, E,
30	A, C, D, F

Potential
max-patterns

- Since BCDE is a max-pattern, no need to check BCD, BDE, CDE in later scan
- R. Bayardo. **Efficiently mining long patterns from databases.** *SIGMOD'98*

PATTERN EVALUATION MEHTODS



INTERESTINGNESS MEASURE

Correlations (Lift)

- play basketball $\Rightarrow$ eat cereal [40%, 66.7%] is misleading
 - The overall % of students eating cereal is 75% > 66.7%.
- play basketball $\Rightarrow$ not eat cereal [20%, 33.3%] is more accurate, although with lower support and confidence
- Measure of dependent/correlated events: [lift](#)

$$lift = \frac{P(A \cup B)}{P(A)P(B)}$$

$$lift(B, C) = \frac{2000 / 5000}{3000 / 5000 * 3750 / 5000} = 0.89$$

$$lift(B, \neg C) = \frac{1000 / 5000}{3000 / 5000 * 1250 / 5000} = 1.33$$

	Basketball	Not basketball	Sum (row)
Cereal	2000	1750	3750
Not cereal	1000	250	1250
Sum(col.)	3000	2000	5000



INTERESTINGNESS MEASURE (2)

Correlations (Lift & χ^2)

- “Buy walnuts $\Rightarrow$ buy milk [1%, 80%]” is misleading if 85% of customers buy milk
- Support and confidence are not good to indicate correlations
- Over 20 interestingness measures have been proposed (see Tan, Kumar, Sritastava @KDD’02)

symbol	measure	range	formula
ϕ	ϕ -coefficient	-1 ... 1	$\frac{P(A,B) - P(A)P(B)}{\sqrt{P(A)P(B)(1-P(A))(1-P(B))}}$
Q	Yule’s Q	-1 ... 1	$\frac{P(A,B)P(\bar{A},\bar{B}) - P(A,\bar{B})P(\bar{A},B)}{P(A,B)P(\bar{A},\bar{B}) + P(A,\bar{B})P(\bar{A},B)}$
Y	Yule’s Y	-1 ... 1	$\frac{\sqrt{P(A,B)P(\bar{A},\bar{B})} - \sqrt{P(A,\bar{B})P(\bar{A},B)}}{\sqrt{P(A,B)P(\bar{A},\bar{B})} + \sqrt{P(A,\bar{B})P(\bar{A},B)}}$
k	Cohen’s	-1 ... 1	$\frac{P(A,B) + P(\bar{A},\bar{B}) - P(A)P(B) - P(\bar{A})P(\bar{B})}{1 - P(A)P(B) - P(\bar{A})P(\bar{B})}$
PS	Piatetsky-Shapiro’s	-0.25 ... 0.25	$P(A,B) - P(A)P(B)$
F	Certainty factor	-1 ... 1	$\max\left(\frac{P(B A) - P(B)}{1 - P(B)}, \frac{P(A B) - P(A)}{1 - P(A)}\right)$
AV	added value	-0.5 ... 1	$\max(P(B A) - P(B), P(A B) - P(A))$
K	Klosgen’s Q	-0.33 ... 0.38	$\sqrt{P(A,B)} \max(P(B A) - P(B), P(A B) - P(A))$
g	Goodman-kruskal’s	0 ... 1	$\frac{\sum_j \max_k P(A_j, B_k) + \sum_k \max_j P(A_j, B_k) - \max_j P(A_j) - \max_k P(B_k)}{2 - \max_j P(A_j) - \max_k P(B_k)}$
M	Mutual Information	0 ... 1	$\frac{\sum_i \sum_j P(A_i, B_j) \log \frac{P(A_i, B_j)}{P(A_i)P(B_j)}}{\min(-\sum_i P(A_i) \log P(A_i) \log P(A_i), -\sum_i P(B_i) \log P(B_i) \log P(B_i))}$
J	J-Measure	0 ... 1	$\max(P(A,B) \log\left(\frac{P(B A)}{P(B)}\right) + P(A\bar{B}) \log\left(\frac{P(\bar{B} A)}{P(\bar{B})}\right), \\ P(A,B) \log\left(\frac{P(A B)}{P(A)}\right) + P(\bar{A}B) \log\left(\frac{P(A \bar{B})}{P(A)}\right))$
G	Gini index	0 ... 1	$\max(P(A)[P(B A)^2 + P(\bar{B} A)^2] + P(\bar{A}[P(B \bar{A})^2 + P(\bar{B} \bar{A})^2] - P(B)^2 - P(\bar{B})^2, \\ P(B)[P(A B)^2 + P(\bar{A} B)^2] + P(\bar{B}[P(A \bar{B})^2 + P(\bar{A} \bar{B})^2] - P(A)^2 - P(\bar{A})^2)$
s	support	0 ... 1	$P(A,B)$
c	confidence	0 ... 1	$\max(P(B A), P(A B))$
L	Laplace	0 ... 1	$\max\left(\frac{NP(A,B)+1}{NP(A)+2}, \frac{NP(A,B)+1}{NP(B)+2}\right)$
IS	Cosine	0 ... 1	$\frac{P(A,B)}{\sqrt{P(A)P(B)}}$
γ	coherence(Jaccard)	0 ... 1	$\frac{P(A,B)}{P(A)+P(B)-P(A,B)}$
α	all_confidence	0 ... 1	$\frac{\max(P(A), P(B))}{P(A,B)}$
o	odds ratio	0 ... ∞	$\frac{P(A,B)P(\bar{A},\bar{B})}{P(\bar{A},B)P(A,\bar{B})}$
V	Conviction	0.5 ... ∞	$\max\left(\frac{P(A)P(\bar{B})}{P(A\bar{B})}, \frac{P(B)P(\bar{A})}{P(\bar{A}B)}\right)$
λ	lift	0 ... ∞	$\frac{P(A,B)}{P(A)P(B)}$
S	Collective strength	0 ... ∞	$\frac{P(A,B) + P(\bar{A}\bar{B})}{P(A)P(B) + P(\bar{A})P(\bar{B})} \times \frac{1 - P(A)P(B) - P(\bar{A})P(\bar{B})}{1 - P(A,B) - P(\bar{A}\bar{B})}$
χ^2	χ^2	0 ... ∞	$\sum_i \frac{(P(A_i) - E_i)^2}{E_i}$



INTERESTINGNESS MEASURE (3)

Null-Invariant

Table 6: Properties of interestingness measures. Note that none of the measures satisfies all the properties.

Symbol	Measure	Range	P1	P2	P3	O1	O2	O3	O3'	O4
ϕ	ϕ -coefficient	$-1 \dots 0 \dots 1$	Yes	Yes	Yes	Yes	No	Yes	Yes	No
λ	Goodman-Kruskal's	$0 \dots 1$	Yes	No	No	Yes	No	No*	Yes	No
α	odds ratio	$0 \dots 1 \dots \infty$	Yes*	Yes	Yes	Yes	Yes	Yes*	Yes	No
Q	Yule's Q	$-1 \dots 0 \dots 1$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Y	Yule's Y	$-1 \dots 0 \dots 1$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
κ	Cohen's	$-1 \dots 0 \dots 1$	Yes	Yes	Yes	Yes	No	No	Yes	No
M	Mutual Information	$0 \dots 1$	Yes	Yes	Yes	No**	No	No*	Yes	No
J	J-Measure	$0 \dots 1$	Yes	No	No	No**	No	No	No	No
G	Gini index	$0 \dots 1$	Yes	No	No	No**	No	No*	Yes	No
s	Support	$0 \dots 1$	No	Yes	No	Yes	No	No	No	No
c	Confidence	$0 \dots 1$	No	Yes	No	No**	No	No	No	Yes
L	Laplace	$0 \dots 1$	No	Yes	No	No**	No	No	No	No
V	Conviction	$0.5 \dots 1 \dots \infty$	No	Yes	No	No**	No	No	Yes	No
I	Interest	$0 \dots 1 \dots \infty$	Yes*	Yes	Yes	Yes	No	No	No	No
IS	Cosine	$0 \dots \sqrt{P(A, B)} \dots 1$	No	Yes	Yes	Yes	No	No	No	Yes
PS	Piatetsky-Shapiro's	$-0.25 \dots 0 \dots 0.25$	Yes	Yes	Yes	Yes	No	Yes	Yes	No
F	Certainty factor	$-1 \dots 0 \dots 1$	Yes	Yes	Yes	No**	No	No	Yes	No
AV	Added value	$-0.5 \dots 0 \dots 1$	Yes	Yes	Yes	No**	No	No	No	No
S	Collective strength	$0 \dots 1 \dots \infty$	No	Yes	Yes	Yes	No	Yes*	Yes	No
ζ	Jaccard	$0 \dots 1$	No	Yes	Yes	Yes	No	No	No	Yes
K	Klosgen's	$(\frac{2}{\sqrt{3}} - 1)^{1/2} [2 - \sqrt{3} - \frac{1}{\sqrt{3}}] \dots 0 \dots \frac{2}{3\sqrt{3}}$	Yes	Yes	Yes	No**	No	No	No	No

where: P1: $O(M) = 0$ if $\det(M) = 0$, i.e., whenever A and B are statistically independent.

P2: $O(M_2) > O(M_1)$ if $M_2 = M_1 + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$.

P3: $O(M_2) < O(M_1)$ if $M_2 = M_1 + \begin{bmatrix} 0 & k \\ 0 & -k \end{bmatrix}$ or $M_2 = M_1 + \begin{bmatrix} 0 & 0 \\ k & -k \end{bmatrix}$.

O1: Property 1: Symmetry under variable permutation.

O2: Property 2: Row and Column scaling invariance.

O3: Property 3: Antisymmetry under row or column permutation.

O3': Property 4: Inversion invariance.

O4: Property 5: Null invariance.

Yes*: Yes if measure is normalized.

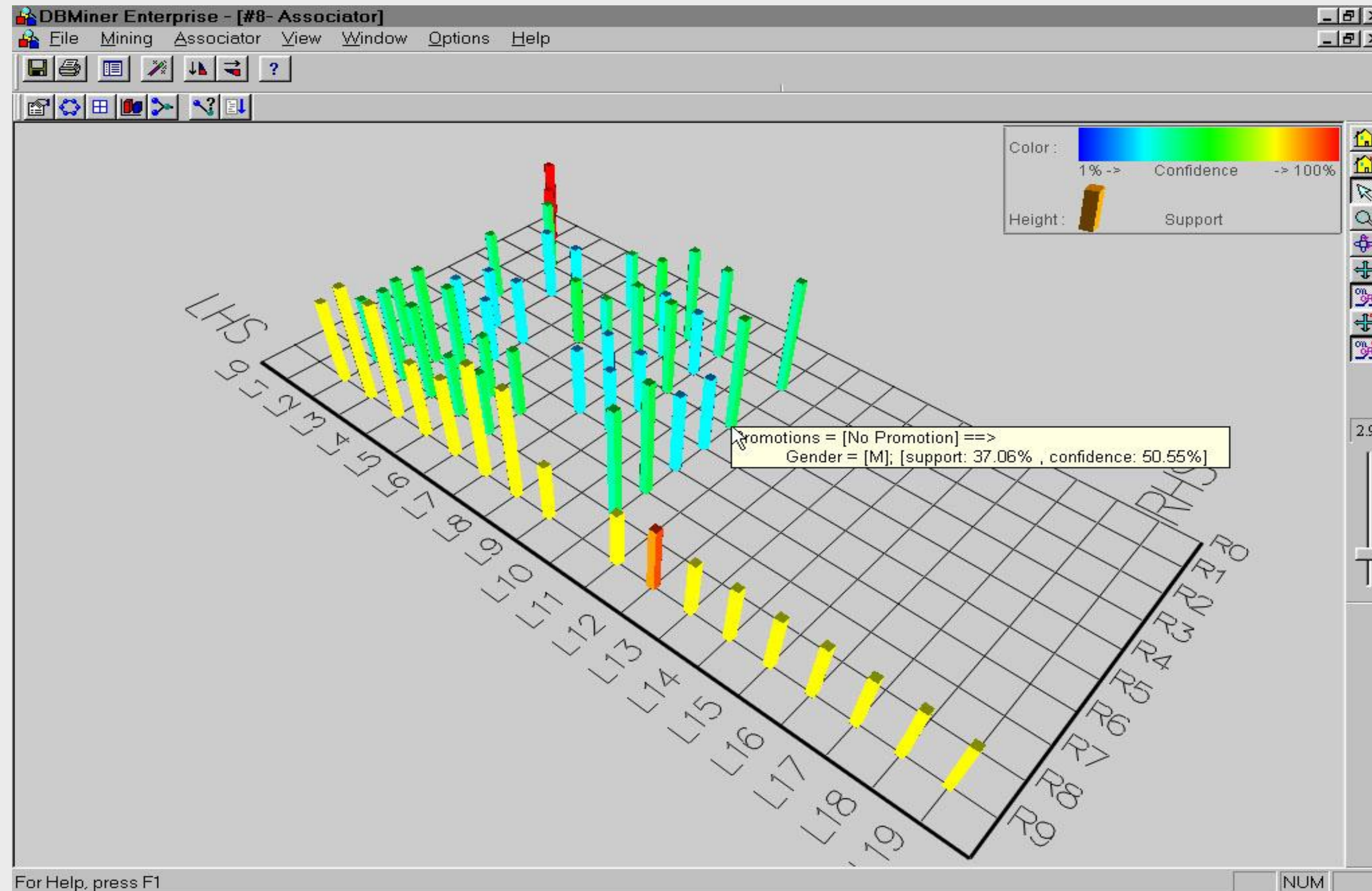
No*: Symmetry under row or column permutation.

No**: No unless the measure is symmetrized by taking $\max(M(A, B), M(B, A))$.



VISUALIZATION OF ASSOCIATION RULES

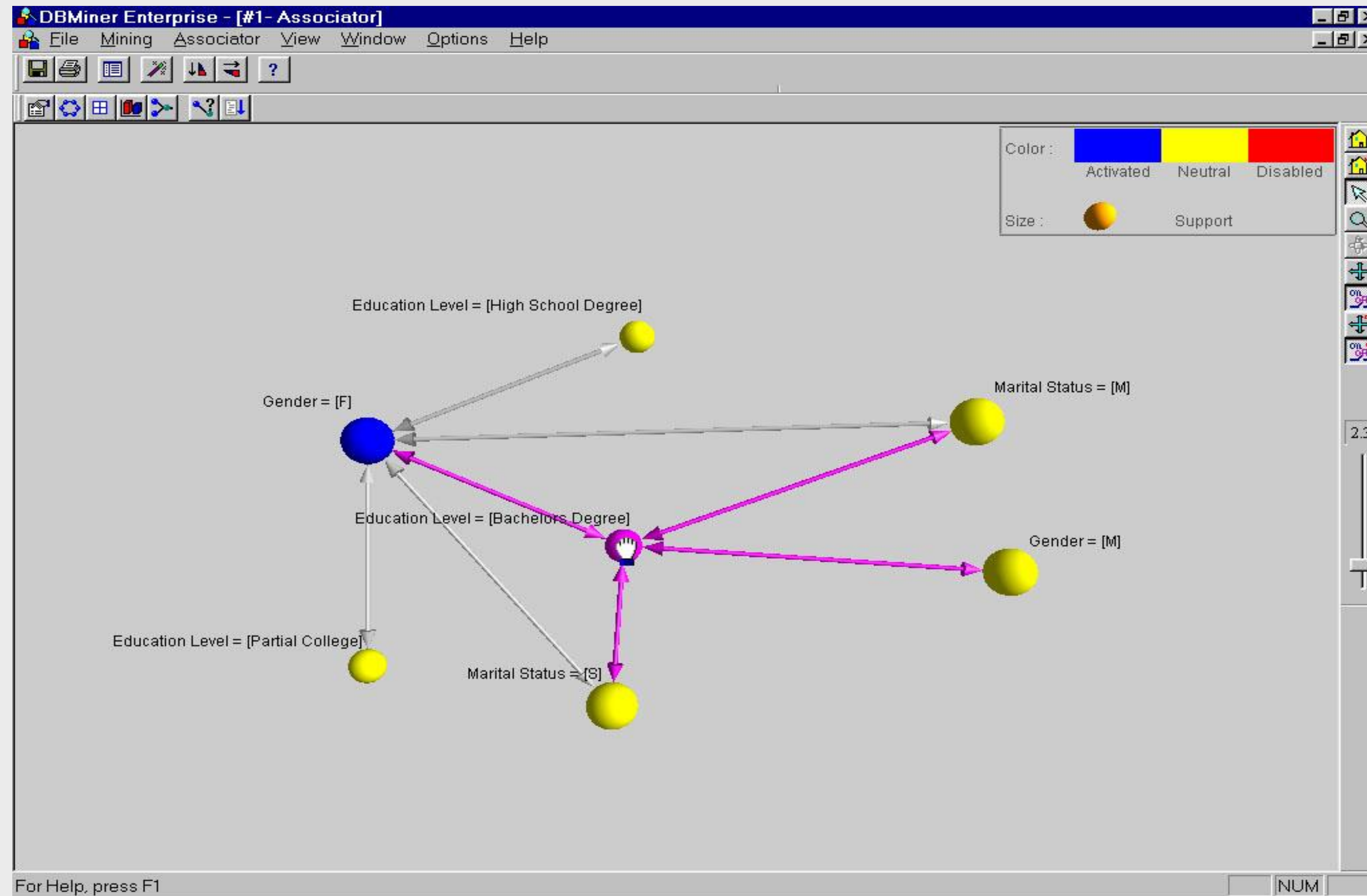
Plane Graph





VISUALIZATION OF ASSOCIATION RULES (2)

Rule Graph



SUMMARY

- Scalable frequent pattern mining methods
 - Projection-based (FPgrowth, CLOSET+, ...)
 - Vertical format approach (ECLAT, CHARM, ...)
- Pattern evaluation methods

EXERCISE

1. Compare the Apriori and FP Growth mechanisms then make a comparative analysis of the a priori and FP growth algorithms.

REFERENCES

Han, J., Kamber, M., & Pei, J. (2023). Data Mining: Concepts and Techniques (4rd ed.). San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.